

Krish Jeswal

Full-Stack Developer & ML Scientist | Electronics & Telecom Undergrad
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Summary

Versatile engineer with a proven track record of shipping real products, from production React/TypeScript web apps with serverless backends to ML pipelines and embedded systems using ESP32 and Python. Active in hackathons, CTFs, and developer communities, with experience leading dev teams. Comfortable across the full stack, ML/data workflows, and the hardware-software boundary, bringing both systems thinking and fast product delivery to whatever I build.

Education

R.V. College of Engineering	Undergraduate
Electronics and Telecommunication	Bengaluru, India • 2024 - 2028
Amity International School	Higher Secondary
Physics, Chemistry, Maths and Computer Science	Ghaziabad, India • 2009 - 2023

Skills

Programming Languages	Frontend	Backend
Python, C, JavaScript, TypeScript, Verilog	React, Tailwind CSS, Vite, Three.js	Node.js, EXPRESS, REST API, Flask, Docker
Databases & Tools	ML/Data	Electronics
PostgreSQL, Supabase, Firebase, Git	NumPy, Pandas, sci-kit learn, OpenCV, Streamlit	Arduino, ESP32, FPGA, UART, LTSpice, KiCAD

Projects

CipherTrace — ML Side-Channel Attack on Masked AES-128

- Built a profiling attack pipeline on the ASCAD dataset (ANSSI, 50k traces, ATmega8515) targeting AES-128 under first-order Boolean masking; evaluated 72 configurations across 6 classifiers × 3 POI strategies using Guessing Entropy as the primary metric.
- Best result: MLP + PCA achieves GE = 16.36 at 500 traces; Identity labels (256-class) reduce GE to 0.46 versus 12.03 for standard Hamming Weight labels, overturning a core SCA assumption. SHAP analysis reveals 2/10 overlap between model-identified and SNR-selected leakage regions.

Intelligent Excel Parser

- Built a **FastAPI** service using **Google Gemini** (gemini-2.5-flash-lite) to intelligently map inconsistent .xlsx column headers to a canonical registry, with automatic fallback to **SequenceMatcher** similarity scoring.
- Engineered a modular pipeline with **openpyxl**, **Pydantic v2** schema validation, and multi-format value parsing; returns structured **JSON** with per-cell confidence levels via an interactive web dashboard.

MoodVue

- Extended a real-time emotion detection app, integrating **DeepFace + OpenCV** via an async **Flask** backend to analyze live webcam feeds and track mood patterns over time.
- Built mood history visualization with Recharts in **React 18 + TypeScript**; integrated **Supabase** for user authentication and data persistence; containerized with **Docker** and deployed on **Vercel + Render**.

Experience

Google Developer Groups - Research & Projects	Jun 2025 – Present
Completed Google Cloud Study Jams, building practical skills across core GCP services including compute, storage, and ML tooling. Co-volunteered at the Vision 2047 Hackathon, supporting event coordination and engaging with teams working on socially impactful tech solutions.	
Coding Club, RVCE - Member	Dec 2024 – Present
Strengthened CS fundamentals through regular collaborative problem-solving sessions focused on algorithms, data structures, and competitive programming. Actively participated in and volunteered at multiple hackathons and CTFs, gaining hands-on experience with time-constrained problem solving and team coordination.	
The Institution of Electronics and Telecommunication Engineers (IETE) - Member	Nov 2025 – Present
Led the development team for the official Student Forum website, overseeing architecture decisions, task delegation, and end-to-end delivery. Contributed to broader technical initiatives within the chapter, including hackathon-style projects and cross-team collaborations on engineering challenges.	